

Homework 09: NumPY and Matplotlib

EEC 3150

For each Python exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

HW<assignment number>_Ex<exercise
number>_<FirstInitial><LastName>.py

Example: HW01_Ex01_GWest.py

For short answer or math exercises, either do your work on a sheet of paper and upload a picture of it to Blackboard or submit some kind of text file or Word document to Blackboard. You can place all such exercises in a single document as long as you clearly label each exercise. These files should follow a similar naming convention as the Python scripts.

Exercise 1 (10 points, Python)

Create and print the following array WITHOUT directly coding any of the elements (only use numpy functions or operations):

```
[[ 1  3  5  7 ]
 [ 9 11 13 15 ]
 [17 19 21 23 ]
 [25 27 29 31 ]]
```

Exercise 2 (10 points, Python)

1. Write a function `Boundary(n, value)` which will create an $n \times n$ matrix in which the elements on the top, bottom, left, and right boundaries are equal to `value` while the inside elements are equal to 0.
2. Demonstrate your function by calling it and printing the output

Here is an example: `Boundary(4, 1)` should produce:

```
[[ 1  1  1  1 ]
 [ 1  0  0  1 ]
 [ 1  0  0  1 ]
 [ 1  1  1  1 ]]
```

Exercise 3 (10 points, Python)

1. Write a function `Truncate(arr)` which takes an array `arr` of integers of any shape and returns a new array with the following properties:
 - Set all values where `arr` is even to 0
 - Set all values where `arr` is odd to 1
2. Demonstrate your function by generating an array of random integers on calling your function on it. Print the output.

Here is an example:

`Truncate(np.array([-1, 2, 11, -4]))`
should produce:

```
[1 0 1 0]
```

Exercise 4 (20 points, Python)

Plot a function of your choice on a domain of your choice.

1. Use `np.linspace` to create an array of x values (you choose the parameters)
2. Create an array of y values by evaluating some function of your choice on the array of x values
3. Plot the data in two ways on the same figure:
1) a green dashed line and 2) red dots. Set the `markersize` and `linewidth` to something other than the default

Exercise 5 (20 points, Python)

The exercise will have you generate random data, plot it, and index it.

1. Generate random data
 - Create 1D `x`, `y` arrays of 1000 random numbers between -2 and 2
2. Plot the data
 - Plot the `x`, `y` arrays as blue dots
3. Index the data
 - Through whatever means you choose, create `x2`, `y2` arrays which only have the points where $-1.5 < x < 1$ and $-1 < y < 0.9$
4. Plot the new data
 - Plot the `x2`, `y2` arrays on the same plot as red x's with an increased `markersize` for readability

More problems on next page...

Exercise 6 (20 points, Python)

1. Create a function to generate a 2D rotation matrix (2D array) (NOTE: sin and cos take radians). For reference, here is the matrix:

$$\begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

2. Create a rotation matrix R with $\theta = 2\pi/n$ where n is an integer variable of your choice (it should be $10 \leq n \leq 100$)
3. Create the vector (1D array) $x = [1 \ 0]$
4. Create an array of vectors (2D array) by iteratively applying the rotation matrix to the vector x . This should produce the points on the unit circle
5. Plot the points